

Build & Evaluate a Linear Regression Model (House Price Predictor)

Objective

The objective of this project is to understand the Machine Learning workflow including data loading, data exploration, preprocessing, model training, prediction, and evaluation using a Linear Regression model on the California Housing dataset.

Dataset Used

The California Housing dataset from scikit-learn was used in this project. The dataset contains housing-related information such as median income, house age, average rooms, average bedrooms, population, occupancy, latitude, longitude, and house price.

Tools & Libraries Used

Python, Pandas, NumPy, Matplotlib, Seaborn, Scikit-learn, Jupyter Notebook

Exploratory Data Analysis (EDA)

The dataset contains 20,640 rows and 9 columns. No missing values were found in the dataset, making it clean and suitable for machine learning. All columns contain numerical values. A correlation heatmap was used to understand relationships between variables and house prices.

Model Training

The dataset was divided into training and testing data using an 80:20 ratio. A Linear Regression model was trained using the training dataset to learn relationships between housing features and house prices.

Prediction

The trained model was used to predict house prices on test data. Predicted values were generated successfully.

Model Evaluation

Results: MAE = 0.5332, RMSE = 0.7456, R^2 Score = 0.5758. The model achieved moderate performance and was able to predict house prices reasonably well.

Conclusion

In this project, a Linear Regression model was built and evaluated using the California Housing dataset. The workflow included dataset loading, exploratory data analysis, model training, prediction, and evaluation. The model achieved moderate performance with an R^2 score of 57.58%. Future improvements can be made using feature engineering, data preprocessing, and advanced machine learning algorithms.